Dr. Konstantin Klemmer

Assistant Professor @ UCL
Principal AI Research Engineer @ LGND AI

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Education

Jul 2018 – Jun 2022 | **Center for Urban Science + Progress, New York University, Brooklyn, USA**
Visiting PhD Student
- Research emphasis: Novel methods for identification of and learning from spatial patterns in urban data.
- Supervision: Professor Daniel B. Neill

Oct 2017 – Jun 2022 | **Department of Computer Science, University of Warwick, Coventry, UK**
PhD in Urban Science
- Dissertation: Improving Neural Networks for Geospatial Applications with Geographic Context Embeddings.
- Supervision: Professor Stephen A. Jarvis, Professor Daniel B. Neill, Dr. Hongkai Wen

Nov 2020 – Apr 2021 | **AI4EO Future Lab, Technical University of Munich & DLR, Munich, Germany**
Beyond Fellow
- Research emphasis: Geospatial machine learning and remote sensing for Earth observation.
- Supervision: Professor Xiaoxiang Zhu

Oct 2018 – Sep 2019 | **The Alan Turing Institute, London, UK**
Enrichment Student
- Research emphasis: Machine learning for urban analytics.

Oct 2016 – Sep 2017 | **Imperial College London & University College London, London, UK**
MSc Transport & Business Management
- Dissertation: Isolating the effect of cycling on local business environments in cities (Grade: 74/100)
- Overall Grade: 72/100 (Distinction)

Oct 2012 – Feb 2016 | **University of Freiburg, Freiburg, Germany**
BSc Economics
- Dissertation: Exploring the spatio-temporal dynamics of carsharing utilization: A case study from Amsterdam (Grade: 1.0)
- Overall Grade: 1.7 (Ranked 4th out of 74 in class)

Work Experience

Sep 2026 – Ongoing | **University College London, UK**
Assistant Professor (Lecturer), Geospatial AI
- Responsibilities: Assistant Professor (Lecturer) in Geospatial AI at UCL Civil, Environmental and Geomatic Engineering and the UCL AI Center.

Aug 2025 – Ongoing | **LGND AI, San Francisco, USA**
Principal AI Research Engineer
- Responsibilities: Bridging research and product development in AI for Earth applications

Dec 2024 – Jul 2025 | **Microsoft Research, Redmond, USA**
Senior Applied Research Scientist
- Responsibilities: Conceptualizing and executing machine learning research projects for industry-relevant applications.

Sep 2022 – Jun 2024 | **Microsoft Research New England, Cambridge, USA**
Postdoctoral Researcher
- Responsibilities: Two-year research fellowship as part of the "ML & Stats" postdoc program.

Jan 2022 – Jun 2022 | **Department of Computer Science, University of Warwick, Coventry, UK**
EPSRC Mathematical Science Fellow
- Responsibilities: Continuation of research conducted throughout doctoral studies.

Nov 2021 – Apr 2022 | **Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), Bonn, Germany**
Expert Consultant
- Responsibilities: AI expert consultant on the appraisal mission of the "Fair Forward" initiative, providing feedback on a large funding scheme for the use of AI in sustainability and development applications in low-income countries.

May 2016 – Dec 2019 | **Geospin GmbH, Freiburg, Germany**
Student Fellow
- Responsibilities: Student fellow in a geospatial data analytics startup. Tasks included data science, social media strategy, market research and business development.
- Manager: Dr. Sebastian Wagner (CEO), Dr. Christoph Gebele

May 2014 – Sep 2016 | **Chair of Economic Policy and Constitutional Economics, University of Freiburg, Freiburg, Germany**
Student Assistant
- Responsibilities: Research assistance (econometric modelling, data preparation), administration of the Chair's website and e-learning modules, event management.
- Manager: Professor Lars P. Feld

Feb 2013 – Apr 2016 | **Chair of Information Systems Research , University of Freiburg, Freiburg, Germany**
Student Assistant
- Responsibilities: Research assistance (data preparation and statistical analysis, data visualization), administration of e-learning modules.
- Manager: Professor Dirk Neumann

Mar 2014 - May 2014 | **Walter Eucken Institute, Freiburg, Germany**
Intern
- Responsibilities: Analysis of scientific articles, data processing, preparation of scientific presentations.
- Manager: Professor Lars P. Feld

Sep 2011 – Aug 2012 | **EVBZ Care Center for People with Disabilities, Steinhöring, Germany**
Federal Voluntary Service
- Responsibilities: Assistance and care for multiply disabled people, carpool supervision, warehousing

Teaching Experience

Sep 2026 – Ongoing | **Department of Civil, Environmental & Geomatic Engineering, University of Warwick, Coventry, UK**
Module Coordinator
- Classes: CEGE0048 – Integrated Design (Undergraduate)

Oct 2017 – Sep 2018 | **Department of Computer Science, University of Warwick, Coventry, UK**
Teaching Assistant
- Classes: CS118 – Programming for Computer Scientists (Undergraduate), CS909 – Data Mining (postgraduate)

Mar 2013 – Apr 2016 | **Chair of Information Systems Research, University of Freiburg, Freiburg, Germany**
Teaching Assistant
- Classes: Introduction to Information Systems (Undergraduate)
- Head instructor in the final year.

Volunteering & Mentoring

Aug 2026 – Ongoing | **Global Fishing Watch, Washington D.C., USA**
Member of the Scientific Advisory Board
- Responsibilities: Advising the Global Fishing Watch board of directors, executive team and chief scientist on research, technology and the application of science in ocean management.

Aug 2020 – Ongoing | **Climate Change AI, Pittsburgh, USA**
Chair of the Communications Committee & Member of the Board of Directors (2021-2023)
- Responsibilities: Leadership role in a fast-growing NGO at the intersection of climate change and artificial intelligence. Tasks include organizational oversight, social media, academic outreach and event organization.

Sep 2015 – Sep 2016 | **Aktionskreis Freiburger Schule, Freiburg, Germany**
Honorary Staff Member
- Responsibilities: Public relations, social media, event management.

Oct 2014 – Mar 2015 | **Mentoring Program of the Economics Department, University of Freiburg, Freiburg, Germany**
Mentor
- Responsibilities: Group supervision for first-semester undergraduate students.

Oct 2012 – Sep 2016 | **Student's Union of the Economics Department, University of Freiburg, Freiburg, Germany**
Student Representative
- Responsibilities: Elected student representative in various committees such as faculty council or student parliament. Co-president of the Economics Student Union (Oct 2013 – Oct 2015).

Awards & Scholarships

Nov 2020 – Apr 2021 | **Technical University of Munich & DLR, Munich, Germany**
Beyond Fellowship

Oct 2018 – Sep 2019 | **The Alan Turing Institute, London, UK**
Enrichment Student Scholarship

Oct 2017 – Dec 2021 | **Engineering and Physical Sciences Research Council, London, UK**
PhD Scholarship

Oct 2017 – Dec 2021 | **University of Warwick, Coventry, UK**
PhD Stipend

Talks

- GISRUK 2017 – CDRC Data Challenge, Leicester, UK, April 2018 (contributed talk)
- Data Natives 2018, London, UK, April 2018 (contributed talk)

- WPCCS 2018, Coventry, UK, June 2018 (contributed talk)
- UbiComp 2018 – Workshop "Pervasive Urban Applications", Singapore, October 2018 (contributed talk)
- NeurIPS 2018 – Workshop on Modeling and decision-making in the spatiotemporal domain, Montreal, Canada, December 2018 (contributed talk)
- A blueprint for urban analytics research, The Alan Turing Institute, Newcastle, UK, April 2019 (invited talk)
- ML4SEP Seminar, University of Zurich, Switzerland, January 2020 (invited talk)
- ICLR 2020 – Workshop "Machine Learning in Real Life" (ML-IRL), Addis Ababa, Ethiopia, April 2020 (contributed short talk)
- ICLR 2020 – Workshop "Climate Change AI" – Energy Day, Addis Ababa, Ethiopia, April 2020 (invited short talk)
- AI4EO seminar series, Munich, Germany, November 2020 (invited talk)
- AI4EO seminar series, Munich, Germany, July 2021 (invited talk)
- Royal Geographical Society – Annual Conference, London, UK, September 2021 (contributed talk)
- European Broadcasting Union – Green AI Micro-Workshop, virtual, March 2022 (invited talk)
- TUM.ai – Seminar series, Munich, Germany, July 2022 (invited talk)
- Harvard University – AI-Assisted Decision-Making for Conservation, Cambridge, USA, October 2022 (invited talk)
- University of Southern California – Frontiers of ML and AI, Los Angeles, USA, November 2022 (invited talk)
- TU Berlin – Climate Change & AI Berlin Brandenburg Workshop, virtual, November 2022 (invited talk)
- National Renewable Energy Laboratory – ALIS Seminar, virtual, January 2023 (invited talk)
- Massachusetts Institute of Technology – Slack Lunch Seminar (SLS), Cambridge, USA, March 2023 (invited talk)
- New York University – Machine Learning for Good (ML4G) Lab Seminar, New York, USA, May 2023 (invited talk)
- Microsoft Research – MSR Montreal Seminar, Montreal, Canada, August 2023 (invited talk)
- Mila – Seminar Talk with David Rolnick's research group, Montreal, Canada, August 2023 (invited talk)
- The Alan Turing Institute – Seminar Talk, London, UK, November 2023 (invited talk)
- AGU Annual Meeting – Beyond the Black Box—Advancing Geo-ML by Incorporating Context with Specialized Architectures, Benchmark Datasets, and Tailored Notions of Interpretability, San Francisco, USA, December 2023 (invited talk)
- University of Cambridge – Energy and Environment Group, Cambridge, UK, March 2024 (invited talk)
- Harvard University – Evaluating the Science of Geospatial AI Conference, Cambridge, USA, May 2024 (invited talk)
- Cyber2A – Cyber2A and PDG Webinar, June 2024 (invited talk)
- Copenhagen University – SSL4EO Summer School, Copenhagen, Denmark, July 2024 (invited talk)
- Technical University of Munich – AI4EO Symposium, Munich, Germany, July 2024 (invited talk)
- Cohere For AI – Community Webinar, November 2024 (invited talk)
- LIDS Computing & Sustainability Seminar – MIT, Cambridge, USA, February 2025 (invited talk)
- Workshop on ML for Remote Sensing – ICLR 2025, Singapore, May 2025 (invited talk)
- Workshop AI4C , Como, Italy, June 2025 (invited talk)
- Remote Sensing Seminar Series, Virginia Tech, Blacksburg, USA, February 2026 (invited talk)

Academic Services

Reviewing

- Journals: Nature, Nature Cities, PLOS ONE, Public Transport – Planning and Operations, Computer Vision and Image Understanding, Environmental Data Science, Environment and Planning B: Urban Analytics and City Science, International Journal of Geographic Information Science, Annals of the American Association of Geographers, Violence against Women, Geo-spatial Information Science, ISPRS Journal of Photogrammetry and Remote Sensing
- Conferences: IEEE ITSC, WI, KDD, AAAI, AISTATS, NeurIPS, CVPR, ICML, ICLR

Area Chair & Senior Program Committee

- AAAI'26 Conference - "AI for Social Impact" track
- NeurIPS'26 Conference

Organization

- Applied Machine Learning Days – AI & Cities track, 2019–2021 (Co-organizer)
- ICLR'20 Workshop "Tackling Climate Change with Machine Learning" – Energy day (Co-organizer)

- NeurIPS'20 Workshop "Machine Learning for the Developing World" – Co-organizer and submissions chair
- NeurIPS'21 Workshop "Machine Learning for the Developing World" – Co-organizer and submissions chair
- ICLR'23 Workshop "Tackling Climate Change with Machine Learning" – Lead organizer
- ICLR'25 Workshop "Tackling Climate Change with Machine Learning" – Lead organizer

Languages

German (native language), English (fluent), French (advanced), Spanish (basic)

Software

Operating Systems: Windows OS, Linux OS
Statistics, Data Science & Machine Learning: R, Python (PyTorch), SPSS
Other: QGIS, Microsoft Office, LaTeX, PHP, PostgreSQL, HTML, CSS, PostGIS, Git, Docker

Publications

2026

1. Corley, I. *et al.* MIND the Gap: A Geographic Implicit Neural Representation with Adjustable Spatial Scale. *Preprint* (2026).
2. **Klemmer, K.** *et al.* Earth Embeddings: Towards AI-centric Representations of our Planet. *IEEE Geoscience and Remote Sensing Magazine* (2026).
3. Knoblauch, S. *et al.* Spatial Representation Learning Beyond Pixels: Unifying Raster Data and Vector Semantics for Human-Centric Geospatial Foundation Models. *Preprint* (2026).
4. Rao, A., Rußwurm, M., **Klemmer, K.** & Rolf, E. Measuring the Intrinsic Dimension of Earth Representations. *Proceedings of the International Conference on Learning Representations (ICLR)* (2026).
5. Rao, A. *et al.* Localized, High-resolution Geographic Representations with Slepian Functions. *Proceedings of the International Conference on Machine Learning (ICML)* (2026).
6. Rodriguez-Pardo, C. *et al.* A Blueprint for Integrated Climate Intelligence. *Preprint* (2026).
7. Zheng, Y. *et al.* EarthEmbeddingExplorer: A Web Application for Cross-Modal Retrieval of Global Satellite Images. *Machine Learning for Remote Sensing (ML4RS) Workshop, International Conference on Learning Representations (ICLR)* (2026).

2025

8. **Klemmer, K.**, Rolf, E., Robinson, C., Mackey, L. & Russwurm, M. SatCLIP: Global, General-Purpose Location Embeddings with Satellite Imagery. *Proceedings of the AAAI Conference on Artificial Intelligence* (2025).
9. **Klemmer, K.** *et al.* ELSA: Local spatial autocorrelation of embeddings. *submitted* (2025).
10. Mickisch, D., **Klemmer, K.**, Teng, M. & Rolnick, D. A Joint Space-Time Encoder for Geographic Time-Series Data. *Tackling Climate Change with Machine Learning Workshop, International Conference on Learning Representations (ICLR)* (2025).
11. Mickisch, D. *et al.* Encoding Spatio-temporal Locations with Orthogonal Function Representations. *submitted* (2025).
12. Zhang, H., Betti, L., **Klemmer, K.**, Rolf, E. & Alvarez-Melis, D. OT on the Map: Quantifying Domain Shifts in Geographic Space. *submitted* (2025).
13. Zhang, H., **Klemmer, K.**, Rolf, E. & Alvarez-Melis, D. Predicting out-of-domain performance under geographic distribution shifts. *Tackling Climate Change with Machine Learning Workshop, International Conference on Learning Representations (ICLR)* (2025).

2024

14. **Klemmer, K.** & Rolf, E. Satellite images reveal untracked human activity on the oceans. *Nature* (2024).
15. Koshiyama, A. *et al.* Towards algorithm auditing: a survey on managing legal, ethical and technological risks of AI, ML and associated algorithms. *Royal Society Open Science* (2024).
16. Rolf, E., **Klemmer, K.**, Robinson, C. & Kerner, H. Mission Critical - Satellite Data is a Distinct Modality in Machine Learning. *Proceedings of the International Conference on Machine Learning (ICML) (Spotlight)* (2024).
17. Russwurm, M., **Klemmer, K.**, Rolf, E., Zbinden, R. & Tuia, D. Geographic Location Encoding with Spherical Harmonics and Sinusoidal Representation Networks. *Proceedings of the International Conference on Learning Representations (ICLR) (Spotlight)* (2024).

2023

18. **Klemmer, K.** & Neill, D. B. Benchmarks and best practices for geospatial predictive modeling with graph neural networks. *submitted* (2023).
19. **Klemmer, K.**, Safir, N. & Neill, D. B. Positional Encoder Graph Neural Networks for Geographic Data. *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)* (2023).
20. Weiss, M. *et al.* An Investigation of General-Purpose Neural Models for Low-Shot Adaptation in Geospatial Applications. *submitted* (2023).
21. Xu, L. *et al. Reflections from the Workshop on AI-Assisted Decision Making for Conservation* 2023.

2022

22. Cunningham*, T., **Klemmer***, **K.**, Wen, H. & Ferhatosmanoglu, H. GeoPointGAN: Synthetic Spatial Data with Local Label Differential Privacy. *submitted* (2022).
23. **Klemmer***, **K.**, Xu*, T., Acciaio, B. & Neill, D. B. SPATE-GAN: Improved Generative Modeling of Dynamic Spatio-Temporal Patterns with an Autoregressive Embedding Loss. *Proceedings of the AAAI Conference on Artificial Intelligence* (2022).
24. Reiersen, G. *et al.* ReforesTree: A Dataset for Estimating Tropical Forest Carbon Stock with Deep Learning and Aerial Imagery. *Proceedings of the AAAI Conference on Artificial Intelligence* (2022).

2021

25. **Klemmer, K.** & Neill, D. B. Auxiliary-task learning for geographic data with autoregressive embeddings. *Proceedings of the ACM International Symposium on Advances in Geographic Information Systems (SIGSPATIAL)* (2021).
26. **Klemmer, K.**, Neill, D. B. & Jarvis, S. A. Understanding spatial patterns in rape reporting delays. *Royal Society Open Science* (2021).
27. **Klemmer, K.**, Saha, S., Kahl, M., Xu, T. & Zhu, X. X. Generative modeling of spatio-temporal weather patterns with extreme event conditioning. *AI: Modeling Oceans and Climate Change (AIMOCC) Workshop, International Conference on Learning Representations (ICLR)* (2021).
28. Luo, M., Du, B., **Klemmer, K.**, Zhu, H. & Wen, H. Deployment Optimization for Shared e-Mobility Systems With Multi-Agent Deep Neural Search. *IEEE Transactions on Intelligent Transportation Systems* (2021).

2020

29. **Klemmer, K.**, Yeboah, G., de Albuquerque, J. P. & Jarvis, S. A. Population Mapping in Informal Settlements with High-Resolution Satellite Imagery and Equitable Ground-Truth. *Machine Learning in Real Life (ML-IRL) Workshop, International Conference on Learning Representations (ICLR)* (2020).
30. Luo, M. *et al.* D3P: Data-driven Demand Prediction for Fast Expanding Electric Vehicle Sharing Systems. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (2020).

2019

31. **Klemmer, K.**, Koshiyama, A. & Flennerhag, S. Augmenting correlation structures in spatial data using deep generative models. *arXiv Preprint* (2019).

2018

32. **Klemmer, K.**, Brandt, T. & Jarvis, S. Isolating the effect of cycling on local business environments in London. *PLOS ONE* (2018).
33. Munoz-Mendez, F., Han, K., **Klemmer, K.** & Jarvis, S. Community Structures, Interactions and Dynamics in London's Bicycle Sharing Network. *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (UbiComp)* (2018).

2017

34. Willing, C., **Klemmer, K.**, Brandt, T. & Neumann, D. Moving in time and space – Location intelligence for carsharing decision support. *Decision Support Systems* (2017).

2016

35. **Klemmer, K.**, Willing, C., Wagner, S. & Brandt, T. Explaining Spatio-Temporal Dynamics in Carsharing: A Case Study of Amsterdam. *AMCIS 2016 Proceedings* (2016).

* denotes equal contribution.